Who Moved, and Who Never Left

84-355 Democracy's Data | Migration between the states, ACS 2024

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Everything here runs as given. Your job is to read what comes out and say what it means.

> **Read `migration-brief.Rmd` first.** This lab runs the same numbered steps with the code exposed. The brief tells you what the question is and why it is hard; this shows you where each number came from.

Migration is apportionment

> *"People are voting with their feet."*

Every ten years the House is redistributed among the states according to where people live. **Nobody votes on that redistribution.** It is the accumulated result of millions of private decisions about jobs, rent and family, and it moves seats and electoral votes without an election.

So: who moves, from where, to where — and **how much of what we think we know about it can the data actually support?**

Steps 1–2 — What a migration statistic is, and one state

```
D  <- "data"
fl <- read.csv(file.path(D, "flows.csv"))        # one row per ordered state pair
st <- read.csv(file.path(D, "states.csv"))       # one row per state, plus DC and PR
ar <- read.csv(file.path(D, "arcs.csv"))         # precomputed arc geometry
mo <- read.csv(file.path(D, "mobility.csv"))     # CPS back-series, one row per year
cy <- read.csv(file.path(D, "county_focus.csv")) # one county, close up
mp <- read.csv(file.path(D, "map_states.csv"))   # state outlines, delta-encoded
ins <- read.csv(file.path(D, "map_insets.csv"))  # Alaska / Hawaii / Puerto Rico boxes
mm <- read.csv(file.path(D, "meta.csv"))         # what the build script counted

S <- st[st$is_state, ]        # Puerto Rico is in the flows but is not a state

# Every number in the prose below is pulled out with these four helpers, so
# nothing in this document is typed in by hand. If the data changes, the
```

```r
# sentences change with it.
mv <- function(k) mm$value[mm$key == k]             # one value from meta.csv
mn <- function(k) as.numeric(mv(k))
n  <- function(x) format(round(x), big.mark = ",")  # 254332 -> "254,332"
pc <- function(x, k = 1) formatC(x, format = "f", digits = k)

# the build script chose all three FROM THE DATA, not from a list
FOCUS    <- mv("focus_state")      # largest net interstate flow, either sign
CONTRAST <- mv("contrast_state")   # fewest inflows that clear their own margin
CTY      <- mv("focus_county")     # most movement of any county in FOCUS
NOR      <- nrow(st) - 1           # origins available to any one state

c(state_pairs = nrow(fl), states_and_dc = nrow(S), possible_origins = NOR)
```

```
#>      state_pairs    states_and_dc possible_origins
#>             2652               51               51
```

Every number here comes from one survey question: **“Where did you live one year ago?”** Cross the answer against where the household lives now and you get a 52 × 52 table — 2,652 ordered pairs once the diagonal is dropped.

```r
F1 <- S[S$state == FOCUS, ]
F1[, c("state", "in_est", "in_moe", "out_est", "out_moe", "net", "net_moe")]
```

```
#>        state in_est in_moe out_est out_moe     net  net_moe
#> 5 California 406873  16100  661205   21844 -254332 27136.14
```

```r
# how many margins of error away from zero is that net figure?
round(abs(F1$net) / F1$net_moe, 1)
```

```
#> [1] 9.4
```

California lost 254,332 more people to other states than it gained from them — the largest net domestic outflow in the country, and 9.4 times its own margin of error.

Step 3 — The arc map

Each arc runs between two states’ **population-weighted centres**: the point around which a state’s residents actually are, not the middle of its outline.

Width is the number of people, and nothing else is. Colour says something different: whether the flow is big enough that the survey can tell it apart from zero, which is what Steps 5 and 6 are about.

```r
MW <- mn("frame_w")     # the drawing frame the build script laid the map out in
MH <- mn("frame_h")

# "x0 y0 dx dy ..." -> absolute coordinates
mdec <- function(p) {
  v <- as.integer(strsplit(p, " ", fixed = TRUE)[[1]])
  list(x = cumsum(v[c(TRUE, FALSE)]), y = cumsum(v[c(FALSE, TRUE)]))
```

```r
}

# ---- WIDTH: the only channel carrying the number of people ---------------
# The scale is ABSOLUTE, not per-panel: WLWD line units at WREF people, so the
# same thickness means the same number of movers in every map below and the key
# in the corner can be read off. The transform is a square root, because these
# flows span roughly 300 to 1 -- drawn linearly the small arcs vanish and the
# big ones become blobs.
WREF <- 50000     # a flow of this many people ...
WLWD <- 5.0       # ... is drawn this thick
WMIN <- 0.40      # and nothing thinner than this, or it disappears
awid <- function(e) pmax(WMIN, WLWD * sqrt(e / WREF))
WKEY <- c(1000, 10000, 50000)      # the reference arcs in the corner key

# ---- COLOUR: a different variable, not a second copy of the width --------
ARC <- "#5fb0e5"      # the survey can tell this flow apart from zero
DIM <- "#999999"      # it cannot

arckey <- function(cex = 0.72) {           # drawn with the map's own function
  x2 <- MW - 22; x1 <- x2 - 78; xt <- x1 - 12
  yy <- c(34, 84, 142)
  text(x2, 186, "migrants per arc", col = "#9fb3c8", cex = cex, adj = 1)
  for (i in seq_along(WKEY)) {
    lines(c(x1, x2), rep(yy[i], 2), col = adjustcolor(ARC, 0.85),
          lwd = awid(WKEY[i]))
    text(xt, yy[i], format(WKEY[i], big.mark = ","), col = "#9fb3c8",
         cex = cex, adj = 1)
  }
}

arcpanel <- function(hub, dir, title, grey_insig = FALSE) {
  d <- ar[ar$hub == hub & ar$dir == dir, ]
  d <- d[order(d$est), ]                       # small first, big drawn on top
  par(bg = "#12161c", mar = c(0.2, 0.2, 1.4, 0.2))
  plot(NA, xlim = c(0, MW), ylim = c(0, MH), asp = 1, axes = FALSE, ann = FALSE)
  for (i in seq_len(nrow(mp))) {
    z <- mdec(mp$pts[i])
    polygon(z$x, z$y, col = "#1e2732", border = "#2f3d4d", lwd = 0.4)
  }
  rect(ins$x0, ins$y0, ins$x1, ins$y1, border = "#2f3d4d", lwd = 0.5)
  lw <- awid(d$est)
  tt <- seq(0, 1, length.out = 40)
  for (i in seq_len(nrow(d))) {
    # quadratic Bezier: start, control point, end
    x <- (1-tt)^2 * d$x0[i] + 2*(1-tt)*tt * d$cx[i] + tt^2 * d$x1[i]
    y <- (1-tt)^2 * d$y0[i] + 2*(1-tt)*tt * d$cy[i] + tt^2 * d$y1[i]
    if (!d$sig[i] && grey_insig)     # Step 6 demotes these to dotted hairlines
      lines(x, y, col = adjustcolor(DIM, 0.55), lwd = 0.7, lty = 3)
    else if (!d$sig[i])              # elsewhere they keep their true width
      lines(x, y, col = adjustcolor(DIM, 0.70), lwd = lw[i])
    else
      lines(x, y, col = adjustcolor(ARC, 0.85), lwd = lw[i])
  }
```

```
  arckey()
  points(st$mx, st$my, pch = 19, cex = 0.3, col = "#7f93a8")
  h <- st[st$state == hub, ]
  points(h$mx, h$my, pch = 21, cex = 1.1, col = "#ffffff", bg = "#C41230", lwd = 1)
  mtext(title, side = 3, line = 0.1, col = "#dbe4ee", cex = 0.8, adj = 0.02)
}

par(mfrow = c(2, 1))
arcpanel(FOCUS, "in",  paste("Arriving in", FOCUS))
arcpanel(FOCUS, "out", paste("Leaving", FOCUS))
```


```
# what the square root actually delivers: the spread of the flows arriving in
# FOCUS, and the spread of the widths they are drawn at
aw <- ar[ar$hub == FOCUS & ar$dir == "in", ]
c(people_widest_over_narrowest = round(max(aw$est) / min(aw$est)),
  width_widest_over_narrowest  = round(max(awid(aw$est)) / min(awid(aw$est)), 1))
```

```
#> people_widest_over_narrowest  width_widest_over_narrowest
#>                        305.0                         11.9
```

A 305-to-one spread in people becomes a 11.9-to-one spread in width. That is the compromise: not proportional, but wide enough that the heaviest arcs are unmistakable and the lightest are still on the page.

```
head(fl[order(-fl$est), c("from_state", "to_state", "est", "moe")], 4)
```

```
#>      from_state   to_state   est  moe
#> 2198 California      Texas 77161 9059
#> 1562   New York New Jersey 56799 8244
#> 1433 California     Nevada 53289 6970
#> 106  California    Arizona 52383 6890
```

```
# which two states exchange the most people, counting both directions?
key   <- paste(pmin(fl$from_state, fl$to_state), pmax(fl$from_state, fl$to_state))
gross <- tapply(fl$est, key, sum, na.rm = TRUE)
head(sort(gross, decreasing = TRUE), 3)
```

```
#>    California Texas       Florida Texas New Jersey New York
#>              122608               97478               92801
```

```
# that pair, one direction at a time, heavier direction first
both <- fl[key == names(which.max(gross)), c("from_state", "to_state", "est", "moe")]
both <- both[order(-both$est), ]
both
```

```
#>      from_state   to_state   est  moe
#> 2198 California      Texas 77161 9059
#> 247       Texas California 45447 6640
```

The heaviest arc runs in both directions and it is the same one. California and Texas exchange more people than any other pair of states — 77,161 from California to Texas, and 45,447 back the other way. Net migration is a small difference between two very large numbers.

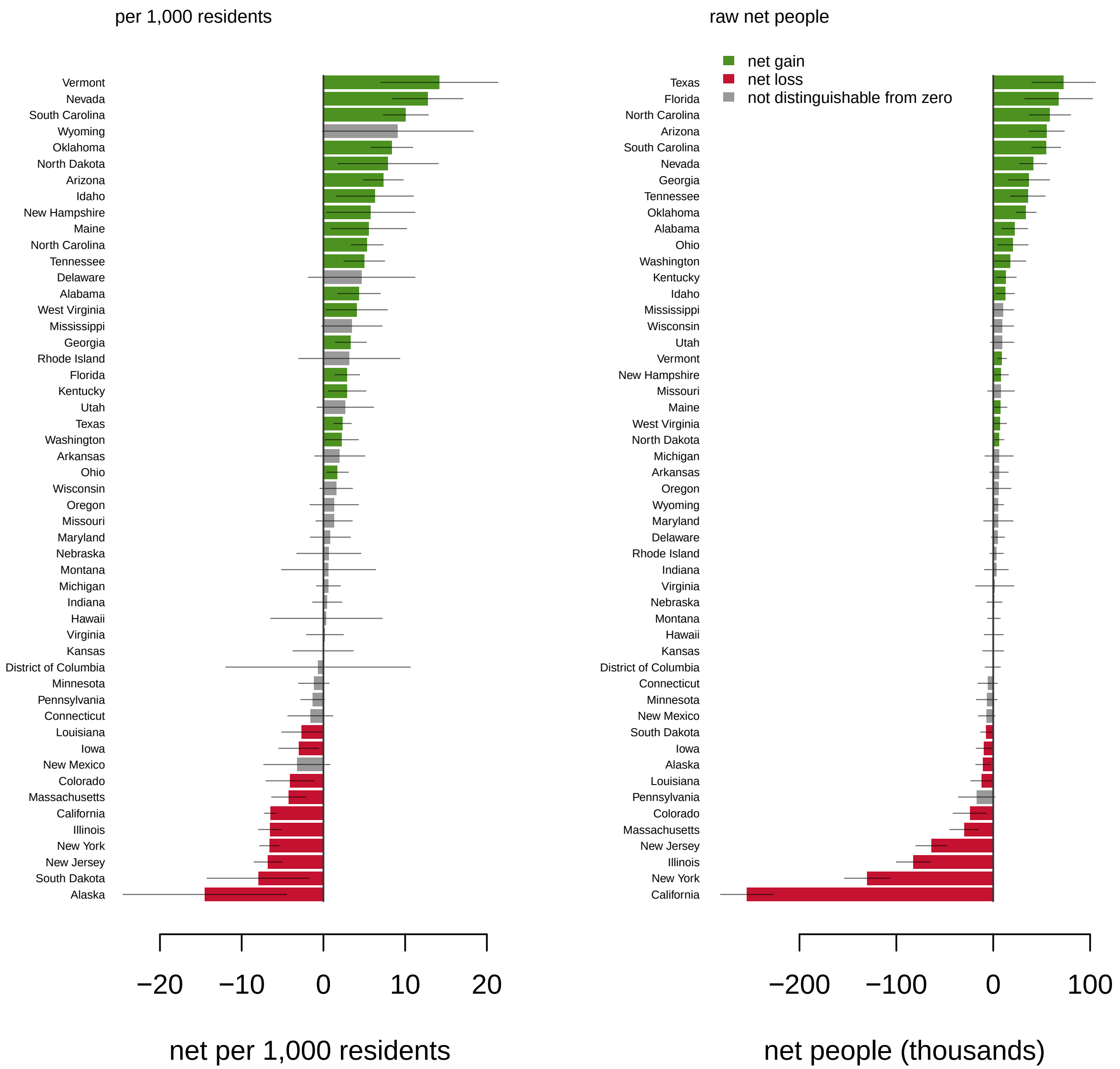
Step 4 — Every state at once

There are two ways to rank states on one chart and they do not agree: **net per 1,000 residents**, which puts Wyoming and Texas on the same scale, and the **raw net count**, which is what moves House seats. Both are drawn below, with the Bureau's 90 percent interval laid over each bar.

```
# the margin of error on a rate is the margin on the count, divided by the same
# population the rate was divided by
S$rate_moe <- 1000 * S$net_moe / S$pop1
S$rate_lo  <- S$net_per1k - S$rate_moe
S$rate_hi  <- S$net_per1k + S$rate_moe
h <- head(S[order(-S$net_per1k), c("state", "net", "net_moe", "net_per1k",
                                   "rate_lo", "rate_hi", "net_sig")], 5)
h[, 3:6] <- round(h[, 3:6], 1)
h
```

```
#>             state   net net_moe net_per1k rate_lo rate_hi net_sig
#> 47        Vermont  9137  4613.9      14.2     7.0    21.3    TRUE
#> 29         Nevada 41278 13985.9      12.8     8.4    17.1    TRUE
#> 42 South Carolina 54676 14936.8      10.1     7.3    12.8    TRUE
#> 52        Wyoming  5301  5375.3       9.1    -0.1    18.3   FALSE
#> 37       Oklahoma 33900 10345.7       8.4     5.8    10.9    TRUE
```

```
divpanel <- function(v, m, ttl, xlab) {
  o <- order(v); vv <- v[o]; mm <- m[o]
  par(mar = c(4.2, 5.0, 1.8, 1.2))
  bp <- barplot(vv, horiz = TRUE, border = NA, names.arg = rep("", length(vv)),
                col = ifelse(!S$net_sig[o], "#999999",
                             ifelse(vv > 0, "#4d9221", "#C41230")),
                xlim = range(c(vv - mm, vv + mm)) * 1.04, xlab = xlab)
  axis(2, at = bp, labels = S$state[o], las = 1, cex.axis = 0.38,
       tick = FALSE, line = -0.7)
  segments(vv - mm, bp, vv + mm, bp, col = "#00000099", lwd = 0.6)  # the interval
  segments(0, min(bp) - 0.5, 0, max(bp) + 0.5, col = "#333", lwd = 1)
  mtext(ttl, side = 3, line = 0.4, cex = 0.7, adj = 0)
}
par(mfrow = c(1, 2))
divpanel(S$net_per1k, S$rate_moe, "per 1,000 residents",
         "net per 1,000 residents")
divpanel(S$net / 1000, S$net_moe / 1000, "raw net people",
         "net people (thousands)")
legend("topleft", bty = "n", cex = 0.55, border = NA,
       fill = c("#4d9221", "#C41230", "#999999"),
       legend = c("net gain", "net loss", "not distinguishable from zero"))
```




```
rbind(head(S[order(-S$net_per1k), c("state", "net", "net_moe", "net_per1k", "net_sig")], 3),
      head(S[order( S$net_per1k), c("state", "net", "net_moe", "net_per1k", "net_sig")], 3))
```

```
#>             state    net   net_moe  net_per1k net_sig
#> 47        Vermont   9137  4613.851  14.182140    TRUE
#> 29         Nevada  41278 13985.937  12.759006    TRUE
#> 42 South Carolina  54676 14936.847  10.087620    TRUE
#> 2          Alaska -10615  7303.921 -14.525953    TRUE
#> 43   South Dakota  -7299  5714.209  -7.979273    TRUE
#> 31     New Jersey -63913 15919.832  -6.798314    TRUE
```

The familiar pattern: the Northeast, the industrial Midwest and California losing; the South and interior West gaining. **But the two panels disagree about who is on top**, and the disagreement is the point:

```r
TOPR <- which.max(S$net_per1k)     # highest rate
TOPN <- which.max(S$net)           # highest raw count
RKN  <- rank(-S$net); RKR <- rank(-S$net_per1k)

# every OTHER jurisdiction whose rate interval reaches into the leader's
OVL <- which(S$rate_hi >= S$rate_lo[TOPR] & seq_len(nrow(S)) != TOPR)

c(rate_leader_rank_on_raw = RKN[TOPR],
  raw_leader_rank_on_rate = RKR[TOPN],
  overlapping_the_leader  = length(OVL),
  spearman = round(cor(S$net, S$net_per1k, method = "spearman"), 2))
```

```
#> rate_leader_rank_on_raw raw_leader_rank_on_rate  overlapping_the_leader
#>                   18.00                   22.00                   17.00
#>                spearman
#>                    0.83
```

Vermont has the highest rate — 14.2 per 1,000 — but that is only 9,137 people, and it ranks 18 of 51 on the raw count. **Texas has the highest raw count**, 72,680 people, and ranks 22 of 51 per resident.

And the per-capita ranking does not survive its own margin of error. Vermont's rate interval runs from 7.0 to 21.3 per 1,000, and **17 other jurisdictions have intervals reaching into it**. The estimate is real; the position is not. Per-capita tables are dominated by small states, where the same absolute uncertainty becomes a much larger relative one.

And look at the grey bars. For 22 of the 51 jurisdictions ranked here — 50 states and the District of Columbia — the net migration figure cannot be told apart from zero.

Steps 5–6 — None of these are counts

> **Before you run this section, look again at the columns you have been ignoring.** Every estimate in this file is printed beside a `moe` — a 90 percent margin of error, published by the Census Bureau itself. There are 2,652 ordered pairs of states.
>
> **Predict: how many describe a flow the survey can actually distinguish from zero?** Write your number down before running the next chunk.

A residual vote cannot be negative; a migration flow cannot be *known* when the estimate is smaller than its own margin of error. That is the test:

```r
# a flow clears zero when the low end of its interval is above zero,
# i.e. when the estimate exceeds its own margin of error
c(pairs         = nrow(fl),
  suppressed_N = sum(is.na(fl$est)),          # "insufficient sample cases"
  exactly_zero = sum(fl$est == 0, na.rm = TRUE),
  distinguishable_from_zero = sum(fl$sig),
  pct_surviving = round(100 * sum(fl$sig) / nrow(fl), 1))
```

```
#>                     pairs              suppressed_N              exactly_zero
#>                    2652.0                     279.0                     276.0
#> distinguishable_from_zero             pct_surviving
#>                    1714.0                      64.6
```

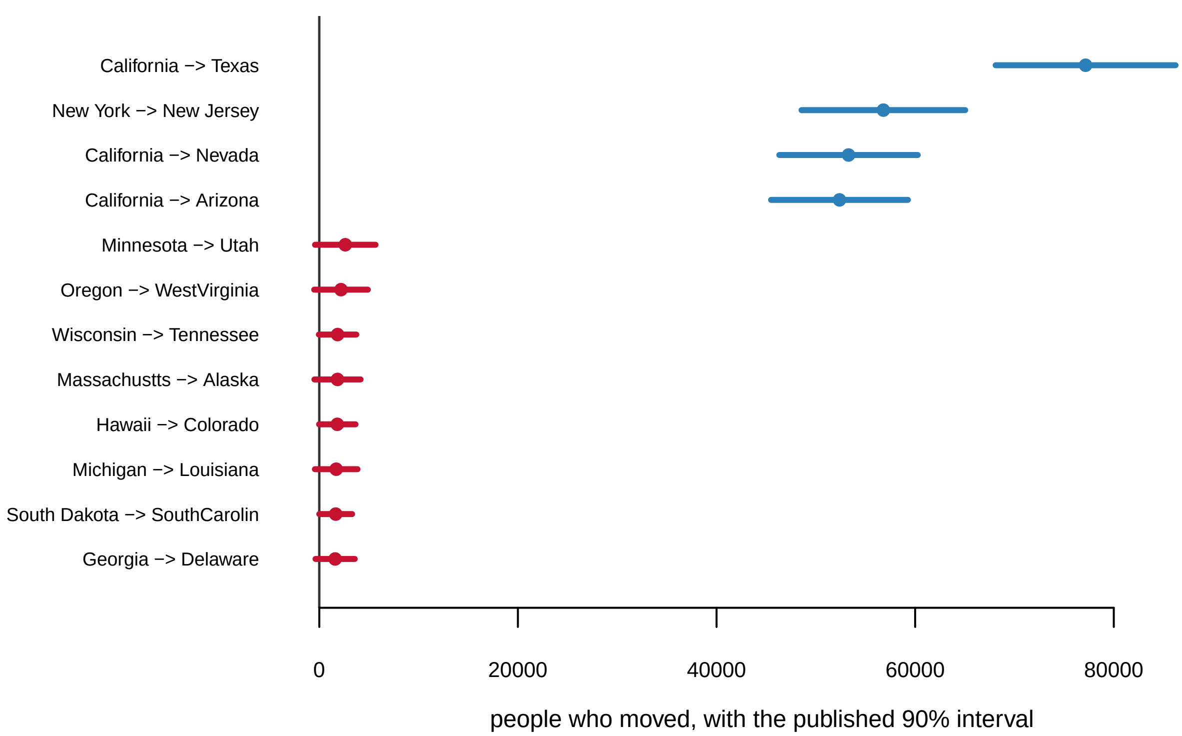
```r
# and the share of actual movers sitting in the pairs that fail
round(100 * sum(fl$est[!fl$sig], na.rm = TRUE) / sum(fl$est, na.rm = TRUE), 2)
```

```
#> [1] 2.21
```

938 of 2,652 state pairs — 35% — do not survive. Those 938 contain 2.21% of the people.

The margin of error does not fall evenly. **It destroys the thin connections and leaves the thick ones untouched** — the median surviving flow is 1,779 people, the median failed flow 129 — which means the arcs that make a flow map *interesting* are exactly the arcs that are not there.

```r
ex  <- fl[!is.na(fl$est), ]; ex <- ex[order(-ex$est), ]
sel <- rbind(head(ex, 4), head(ex[!ex$sig, ], 8))
sel$lab <- paste(abbreviate(sel$from_state, 12), "->", abbreviate(sel$to_state, 12))
sel <- sel[order(sel$est), ]
par(mar = c(3.6, 9.4, 1.0, 1.4))
yy <- seq_len(nrow(sel))
plot(NA, xlim = c(min(0, min(sel$est - sel$moe)), max(sel$est + sel$moe) * 1.04),
     ylim = c(0.4, nrow(sel) + 0.6), axes = FALSE, ann = FALSE)
abline(v = 0, col = "#333", lwd = 1.2)
cl <- ifelse(sel$sig, "#2c7fb8", "#C41230")
segments(sel$est - sel$moe, yy, sel$est + sel$moe, yy, col = cl, lwd = 3)
points(sel$est, yy, pch = 19, cex = 0.8, col = cl)
axis(1, cex.axis = 0.7)
axis(2, at = yy, labels = sel$lab, las = 1, cex.axis = 0.62, tick = FALSE, line = -0.5)
mtext("people who moved, with the published 90% interval", side = 1, line = 2.3, cex = 0.75)
```




```r
# the largest flow in the country that cannot be told apart from zero
bad <- fl[!fl$sig, ][which.max(fl$est[!fl$sig]), ]
bad[, c("from_state", "to_state", "est", "moe")]
```

```
#>      from_state to_state  est  moe
#> 2268  Minnesota     Utah 2617 3046
```

Minnesota to Utah: 2,617 ± 3,046. A published statistic whose own publisher says it is consistent with nobody making that move at all.

And whether your state's map survives this test depends mostly on **how big your state is**:

```r
rbind(head(S[order(-S$sig_in_n), c("state", "pop1", "sig_in_n", "sig_out_n")], 3),
      head(S[order( S$sig_in_n), c("state", "pop1", "sig_in_n", "sig_out_n")], 3))
```

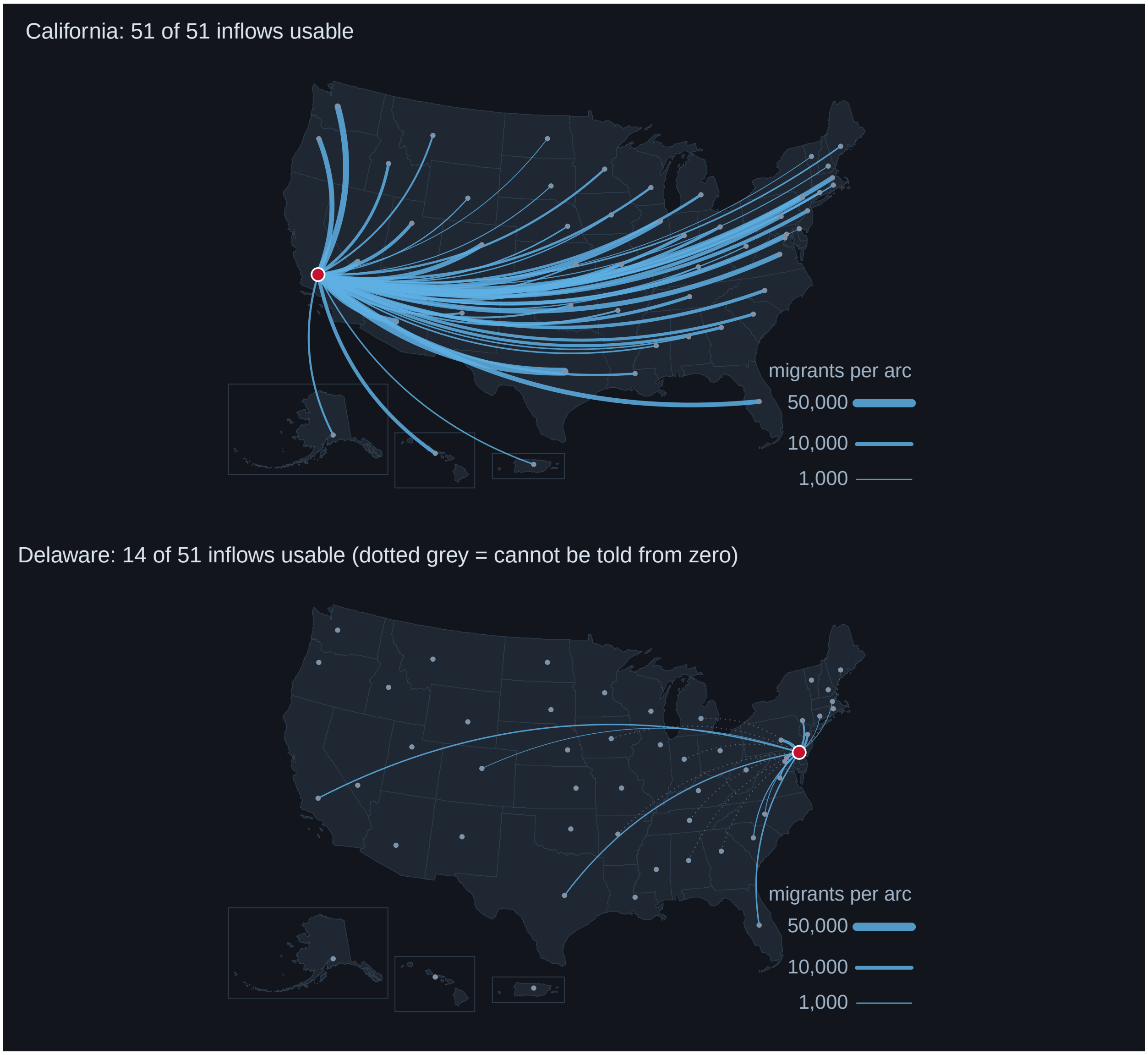
```
#>           state     pop1 sig_in_n sig_out_n
#> 5    California 39039388       51        50
#> 10      Florida 23163294       50        49
#> 45        Texas 30944767       50        51
#> 8      Delaware  1043560       14        13
#> 35 North Dakota   787926       15        16
#> 41 Rhode Island  1103757       15        16
```

```r
round(cor(log(S$pop1), S$sig_in_n), 2)
```

```
#> [1] 0.91
```

```r
C1 <- S[S$state == CONTRAST, ]   # the state with the fewest usable inflows
```

```r
par(mfrow = c(2, 1))
arcpanel(FOCUS, "in",
         sprintf("%s: %d of %d inflows usable", FOCUS, F1$sig_in_n, NOR),
         grey_insig = TRUE)
arcpanel(CONTRAST, "in",
         sprintf("%s: %d of %d inflows usable (dotted grey = cannot be told from zero)",
                 CONTRAST, C1$sig_in_n, NOR),
         grey_insig = TRUE)
```



A flow map of a small state is mostly a picture of the sample size. Delaware is not less connected to the country than California; it is smaller, so 37 of its 51 possible inflows land in too little sample to resolve.

Steps 7–8 — Who a state is made of, and who came from outside

Flows are a year. **Stocks are a century.** The share of a state's residents born in that state records every migration that ever happened to it.

```
o <- S[order(-S$born_state_pct), c("state", "born_state_pct",
                                   "born_otherstate_pct", "born_foreign_pct")]
rbind(head(o, 5), tail(o, 5))
```

```
#>                     state born_state_pct born_otherstate_pct born_foreign_pct
#> 19              Louisiana       77.01995            17.01405         5.823470
#> 23               Michigan       75.11978            16.34704         8.400783
```

```
#> 36                 Ohio       74.27112            19.21251          6.148113
#> 25          Mississippi       70.03566            26.68148          3.147760
#> 39         Pennsylvania       69.59210            20.24140          9.078038
#> 52              Wyoming       40.36279            55.19334          4.248508
#> 3               Arizona       39.14018            45.81843         14.827685
#> 9  District of Columbia       34.79943            47.79181         17.208544
#> 10              Florida       34.73130            38.12741         24.612725
#> 29               Nevada       28.46887            49.83518         21.318777
```

```r
# the national figure, weighted by population rather than averaged over states
round(100 * sum(S$born_state_est) / sum(S$pop_total), 1)
```

```
#> [1] 56.5
```

Louisiana 77.0%. Nevada 28.5%. Nationally 56.5% of Americans live in the state they were born in.

```r
h <- head(S[order(-S$abroad_per1k), ], 6)
h[, c("state", "abroad_est", "abroad_moe", "abroad_per1k", "net")]
```

```
#>                   state abroad_est abroad_moe abroad_per1k    net
#> 9  District of Columbia       9035       2213    13.003405   -455
#> 10              Florida     299482      16581    12.929163  67630
#> 41         Rhode Island      11889       2713    10.771393   3492
#> 49           Washington      82568       9474    10.480935  17707
#> 45                Texas     319607      21877    10.328305  72680
#> 31           New Jersey      90730       9234     9.650792 -63913
```

"From abroad" is a statement about geography, not about citizenship or legal status. It counts everyone who lived outside the 50 states and DC a year ago: returning citizens, military families, students, people arriving from Puerto Rico and the U.S. Island Areas — already citizens and nationals — and immigrants. Nothing in this file separates them further.

Of the 6 states taking the most arrivals from abroad per resident, 2 are **simultaneously losing people to other states** — two populations moving in opposite directions through the same place, which a single "net migration" headline cancels against each other.

And note that **the survey only interviews people who live here**. There is no matching number for people who left the country. Emigration is not small in this dataset; it is *absent*.

Step 9 — One county, and the estimate coming apart

```r
# county_focus.csv holds only the largest inflows, to keep the file small.
# The counts for the WHOLE county were computed by the build script:
head(cy[, c("state_a", "county_a", "into_b", "into_b_moe", "sig_in")], 6)
```

```
#>      state_a              county_a into_b into_b_moe sig_in
#> 1 California         Orange County  38761       2562   TRUE
#> 2 California San Bernardino County  34920       2839   TRUE
#> 3 California      Riverside County  23292       2434   TRUE
#> 4     Nevada          Clark County  12873       1782   TRUE
#> 5 California           Kern County  10503       1253   TRUE
#> 6 California      San Diego County  10436       1312   TRUE
```

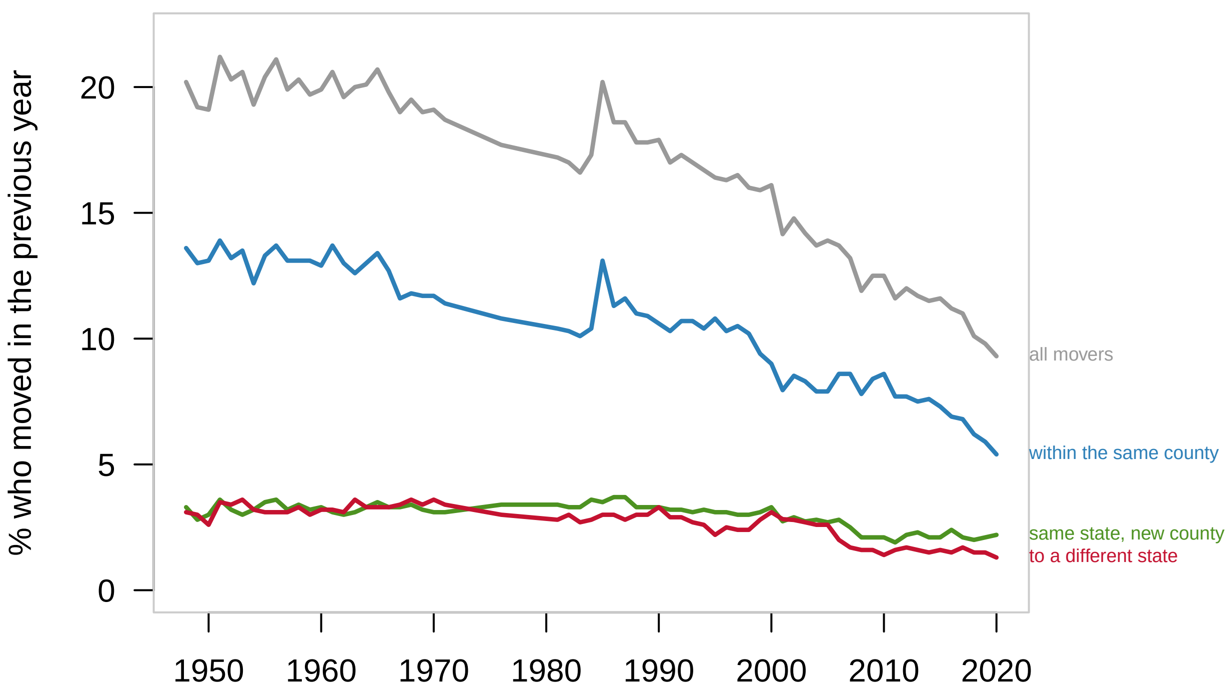
```r
c(county_pairs_reported = mn("county_all_pairs"),
  distinguishable_from_zero = mn("county_all_sig"),
  pct_surviving_county = round(100 * mn("county_all_sig") / mn("county_all_pairs"), 1),
  pct_surviving_state  = mn("pct_sig_of_pairs"))
```

```
#>    county_pairs_reported distinguishable_from_zero      pct_surviving_county
#>                   1234.0                     338.0                      27.4
#>      pct_surviving_state
#>                     64.6
```

338 of 1,234 county pairs feeding Los Angeles County survive — 27.4%, against 64.6% at state level — and that is *after* pooling five years of sample instead of one. County-to-county is a **five-year** estimate (2016-2020) and cannot be put on the same axis as the one-year table used everywhere above.

Step 10 — Americans move less than they used to

```r
par(mar = c(3.8, 4.0, 1.6, 9.2))
plot(NA, xlim = range(mo$year), ylim = c(0, max(mo$movers) * 1.04),
     xlab = "", ylab = "% who moved in the previous year", axes = FALSE)
axis(1); axis(2, las = 1); box(col = "#ccc")
V <- list(c("movers", "#999999", "all movers"),
          c("same_county", "#2c7fb8", "within the same county"),
          c("same_state_diff_county", "#4d9221", "same state, new county"),
          c("diff_state", "#C41230", "to a different state"))
for (v in V) lines(mo$year, mo[[v[1]]], col = v[2], lwd = 2.2)
for (v in V) text(max(mo$year) + 1.2, mo[[v[1]]][nrow(mo)], v[3], pos = 4,
                  cex = 0.6, col = v[2], xpd = NA)
```



```r
f <- mo[1, ]; l <- mo[nrow(mo), ]
rbind(f, l)[, c("year", "movers", "diff_state")]
```

```
#>    year movers diff_state
#> 1  1948   20.2        3.1
#> 65 2020    9.3        1.3
```

```r
# interstate moving now, as a share of what it was in the first year
round(100 * l$diff_state / f$diff_state)
```

```
#> [1] 42
```

In 1948, 20.2% of Americans had moved in the previous year. By 2020 it was 9.3%. Interstate moving fell from 3.1% to 1.3% — 42% of what it was.

This is the **Current Population Survey**, not the ACS — a different survey with a different design, not comparable cell by cell, but the only thing that reaches back that far.

Every argument about Americans sorting themselves across the country is happening against a long decline in Americans moving at all.

Step 11 — What you can now say

```r
g <- head(S[order(-S$net), c("state", "net", "net_moe")], 2)   # biggest gainers
g
```

```
#>      state   net  net_moe
#> 45   Texas 72680 32775.43
#> 10 Florida 67630 34928.61
```

You can say who is gaining and losing: California down 254,332 net to other states, Texas and Florida up 72,680 and 67,630, all several times their own margins of error. And that 56.5% of Americans live in the state they were born in, ranging from 28.5% to 77.0% across states.

You can say something sharper. Of 2,652 state-to-state connections only 1,714 can be distinguished from zero, and those carry 97.8% of the movers. **The uncertainty is concentrated exactly where the interesting-looking pattern is**, so the more distinctive a flow claim sounds, the more likely it is to be an artefact of sample size.

You cannot say why anybody moved — no reason is recorded, so politics cannot be tested here. You cannot say anything about people who left the country. And you cannot call any of this a count of people.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Audit a state's map. Pick any state and count how many of its 51 possible inflows and outflows clear their own margin of error (`S[S$state == "...", c("sig_in_n","sig_out_n")]`). Then say what an arc map of that state would be honest about and what it would fake. Compare a big state to a small one.

Option B — The league table nobody checks. Find the 22 states whose net migration cannot be distinguished from zero (`S[!S$net_sig, ]`). These states appear in "winners and losers" journalism every year. What should a story about them actually say, and how would you write the sentence?

Option C — Born here versus arrived. Compare `born_state_pct` to `net_per1k`. The correlation is weak (r = -0.25) — find the states that break the pattern most and explain what each one is. What is born-in-state share actually measuring, if not this year's migration?

Option D — Your own question. 2,652 state pairs, 51 jurisdictions, a 72-year back-series and one county in close-up.

What this lab cannot tell you

- **Why anybody moved.** No motive is recorded anywhere in this data, so the argument about political sorting cannot be settled or even tested here.
- **Anything about emigration.** The survey interviews people who live in the United States. People who left are not a small number in this file; they are not in it.
- **Anything about legal status.** "From abroad" is a residence category. It mixes returning citizens, movers from U.S. territories, students and immigrants, and the file separates none of them.
- **Whether a small state's flows are real.** For 938 state pairs the honest answer is that the survey does not know, and at county level that is true of 73% of pairs.
- **Anything across the two ACS tables.** One-year and five-year estimates are different instruments over different windows.

On the discussion board

By **Friday, 11:59 p.m.** A sentence or two is enough.

Three that would make good posts:

- A newspaper reports that 2,617 people moved from Minnesota to Utah. The Census Bureau says ± 3,046. What should the newspaper have printed?
- 77% of people in Louisiana were born there; 28% of people in Nevada were. Should we expect those two states to have different kinds of politics, and why?
- Americans move at 46% of the rate they did in 1948. Does that make the "Big Sort" argument more plausible or less?

Where the data came from

State-to-state flows — U.S. Census Bureau, *State-to-State Migration Flows*, **2024 ACS 1-year** estimates, Table T13, with the Bureau's published 90 percent margins of error.

Place of birth — *State of Residence by Place of Birth*, 2024 ACS 1-year, Table T13.

County-to-county flows — *County-to-County Migration Flows*, **2016-2020 ACS 5-year** estimates.

The back-series — *Geographical Mobility*, CPS historical Table A-1, 1948 to 2020. **A different survey from the ACS.**

Map geometry — 2020 Census centers of population by state, and the 2020 cartographic boundary file.

`data/build-data.R` parses all five and writes the CSVs this lab reads. It chooses the focus state, the contrast state and the focus county **from the data** — largest net flow, fewest usable inflows, most movement — rather than from a list.